

WEEK 1 — DATA QUALITY & INTEGRATION REPORT	
Project: Project Foresight	
Scope: Data ingestion, profiling, cleaning, validation, integration and reproducibility	
Final Status: COMPLETED	

SECTION	DETAILS
1. Objective	Ingest all four supplied datasets; profile and validate data quality; apply documented cleaning and transformations; resolve identified data-quality issues; integrate the datasets; and create a reproducible one-command pipeline producing an analysis-ready final dataset.

DATASET	ROWS	INITIAL COLUMNS	FINAL / NOTES
Sales Daily	36,550	6	Validated; used as the sales fact table.
SKU Master	50	8	10 after adding Margin_Percentage and Product_Age_Days.
Calendar	731	11	Validated; daily coverage from 2024-01-01 to 2025-12-31.
Inventory Snapshots	4,800	8	Filtered to 1,200 valid SKU-master-linked rows.

2. SALES DAILY — VALIDATION		
CHECK / FIELD	RESULT	DECISION / INTERPRETATION
Missing values	0 in all columns	No missing-value cleaning required.
Duplicate rows	0	No duplicate rows found.
Date range	2024-01-01 to 2025-12-31	Converted Date to datetime.
Zero Units Sold	202	Retained as valid zero-sales observations.
Negative Units Sold	0	No invalid negative sales quantities.
Zero Revenue	202	Matches the zero-sales records; retained.
Negative Revenue	0	No invalid negative revenue.
Revenue validation	0 mismatches > 0.01	Revenue agrees with Units_Sold × Price.
Promotion values	0: 32,800; 1: 3,750	Only valid binary promotion values found.
Invalid Promotion values	None	No correction required.
Unique SKUs	50	All sales SKUs are represented in SKU Master.
Missing SKUs	0	No missing SKU values.
Duplicate Date + SKU	0	Daily SKU grain is unique.
Sales dates missing from Calendar	0	Full calendar coverage for sales dates.
Promotion=1 without Calendar event	0	Promotion records are consistent with Calendar.

3. SKU MASTER — VALIDATION & DERIVED FEATURES		
CHECK / TRANSFORMATION	RESULT	DECISION / INTERPRETATION
Missing values	0	No missing values.
Duplicate rows	0	No duplicate records.
Duplicate SKUs	0	SKU identifiers are unique.
Launch_Date	Converted to datetime	Required for date-based analysis.
Margin_Percentage	Added	$\text{Gross_Margin_Per_Unit} \div \text{Selling_Price} \times 100$.
Product_Age_Days	Added	Calculated using 2025-12-31 as reference date.
New-column missing values	0	Both derived columns populated.
Negative Product Age	0	No invalid future launch dates.
Margin Percentage range	-486.18% to 96.39%	Negative margins retained as meaningful business conditions.
Selling Price $\leq$ Cost Price	16 SKUs	Retained and documented; potentially loss-making products are analytically meaningful.
Gross Margin mismatches	$0 > 0.01$ tolerance	Gross margin agrees with $\text{Selling_Price} - \text{Cost_Price}$.
Final size	50×10	Cleaned SKU Master saved as sku_master_cleaned.csv.

4. CALENDAR — VALIDATION		
CHECK	RESULT	DECISION / INTERPRETATION
Date coverage	2024-01-01 to 2025-12-31	731 consecutive daily records.
Duplicate dates	0	Each calendar date is unique.
Missing dates	0	Complete daily calendar.
Missing values	Only expected blanks in holiday/promotion_event	Non-event blanks retained as valid representation.
Year consistency	True	Calendar year agrees with date.
Month consistency	True	Calendar month agrees with date.
Weekend consistency	True	is_weekend agrees with date.
Quarter consistency	True	Quarter agrees with month/date.
Day-of-week consistency	True	day_of_week agrees with date.
Season consistency	True	Season agrees with defined month mapping.
Week consistency	True	Week values agree with ISO week.
Holiday consistency	True	Holiday flag and holiday field are consistent.

5. INVENTORY SNAPSHOTS — VALIDATION & SKU RECONCILIATION		
CHECK / ISSUE	RESULT	DECISION / RATIONALE
Missing values	0	No missing values.

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Duplicate rows	0	No duplicate rows.
Negative Current_Stock	0	No invalid negative stock.
Negative On_Order	0	No invalid negative on-order quantity.
Invalid Lead_Time_Days (≤0)	0	All lead times valid.
Negative Safety_Stock	0	No invalid negative safety stock.
Negative Reorder_Point	0	No invalid negative reorder points.
Duplicate Snapshot_Date + SKU	0	Snapshot SKU grain is unique.
Inventory unique SKUs	200	Compared against 50 SKUs in SKU Master.
Unmatched SKUs	150	SKUs not present in SKU Master; documented rather than silently ignored.
Rows from unmatched SKUs	3,600	These rows were excluded from integrated inventory to preserve referential integrity.
Filtered inventory	1,200 rows / 50 SKUs	Only inventory records linked to SKU Master were retained.

6. DATA INTEGRATION RESULTS		
INTEGRATION STAGE	RESULT	VALIDATION
Sales + SKU Master	36,550 × 15	0 missing Product_Name values after merge.
Sales + SKU + Calendar	36,550 × 26	0 missing calendar year values after merge.
Final merge before duplicate-date cleanup	36,550 × 33	Inventory linked on Date + SKU.
Inventory matched	1,200	Matched records had Current_Stock available.
Inventory not matched	35,350	No inventory snapshot existed for those sales Date + SKU combinations; nulls were retained.
Duplicate calendar date column	Removed	Removed duplicate 'date' field after joining on Date/date.
Final integrated dataset	36,550 × 32	Saved as integrated_sales_dataset.csv.

7. CLEANING DECISIONS & RATIONALE		
DATA ISSUE / DECISION	ACTION TAKEN	RATIONALE
String date fields	Converted relevant date fields to datetime.	Reliable date validation, filtering and joins.
Zero-sales records	Retained 202 records.	Zero demand is a valid observation; removing it could distort demand patterns.
16 SKUs with Selling Price ≤ Cost Price	Retained and documented.	Potential loss-making products are useful for later business analysis.
150 inventory SKUs absent from SKU Master	Filtered 3,600 corresponding inventory rows.	Prevents invalid SKU dimension relationships in the integrated dataset.
Calendar non-event blanks	Retained.	Blank holiday/promotion fields represent dates without such events.
Derived SKU features	Added Margin_Percentage and Product_Age_Days.	Creates requested analytical features without altering source facts.

8. REPRODUCIBLE PIPELINE	
PIPELINE ITEM	DETAIL
Pipeline file	pipeline.py
Execution command	python pipeline.py
Execution order	sales_daily.py → sku_master.py → calender.py → inventory_snapshots.py → integration.py
Input	sales_daily.csv, sku_master.csv, calendar.csv, inventory_snapshots.csv
Cleaned outputs	sku_master_cleaned.csv, calendar_cleaned.csv, inventory_snapshots_cleaned.csv
Final output	integrated_sales_dataset.csv
Pipeline test result	Successful — all scripts executed and final file generated.
Reproducibility status	Week 1 requirement satisfied: workflow can be rerun with one command.

WEEK 1 REQUIREMENT	STATUS	EVIDENCE / OUTPUT	FINAL
Ingest all 4 datasets	Complete	All four raw CSVs load successfully.	✓
Profile datasets	Complete	Shapes, columns, types and quality checks completed.	✓
Code cleaning process	Complete	Existing cleaning scripts execute through pipeline.py.	✓
Handle/document data-quality issues	Complete	150 unmatched inventory SKUs and other findings documented.	✓
Integrate datasets	Complete	Final integrated dataset generated at 36,550 × 32.	✓
One-command reproducibility	Complete	python pipeline.py runs the complete Week 1 workflow.	✓
Data-quality report	Complete	This report.	✓
Week 1 overall status	COMPLETE	All identified Week 1 technical and documentation deliverables completed.	✓

FINAL ARTIFACT	STATUS	DESCRIPTION	WEEK 1
sales_daily.py	Ready	Sales profiling and validation script.	✓
sku_master.py	Ready	SKU validation, derived columns and cleaned output.	✓
calender.py	Ready	Calendar validation and cleaned output.	✓
inventory_snapshots.py	Ready	Inventory validation, SKU reconciliation and cleaned output.	✓
integration.py	Ready	Sales, SKU, Calendar and Inventory integration.	✓
pipeline.py	Ready	One-command Week 1 workflow.	✓
integrated_sales_dataset.csv	Generated	Final analysis-ready dataset: 36,550 rows × 32 columns.	✓